Bogyeom Park

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I study how to design human-centered agentic AI systems that help people understand their situation, make meaningful decisions, and take purposeful action. I build the systems I study - multi-agent LLM pipelines, VR tasks in Unity, and multimodal behavioral models - and test them where the judgement carries stakes: tutoring, debate, career counseling, and early cognitive screening. **12 peer-reviewed papers (6 first-authored)**, including a first-authored validation study in *JMIR* (JCR Q1, JIF 8.2); **1 patent filed**; **6 funded projects** since 2021.

EDUCATION

Seoul National University of Science and Technology Seoul, South Korea
Ph.D. in Applied Artificial Intelligence *Mar. 2023 - Present*
- Advisor: Kyoungwon Seo

Seoul National University of Science and Technology Seoul, South Korea
B.S. in Electrical and Information Engineering *Mar. 2019 - Feb. 2023*

HONORS & AWARDS

- AI SeoulTech Graduate Scholarship (KRW 10,000,000), Seoul Scholarship Foundation, 2025
- Best Paper Award (co-author), HCI Korea 2025
- Best Student Paper Bronze Award, IEEE Seoul Section, 2024
- National Science and Engineering Undergraduate Scholarship (full tuition), Korea Student Aid Foundation, 2021-2022
- Best Capstone Design Award, Seoul National University of Science and Technology, 2022

JOURNAL ARTICLES

Integrating Biomarkers From Virtual Reality and Magnetic Resonance Imaging for the Early Detection of Mild Cognitive Impairment Using a Multimodal Learning Approach: Validation Study
Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo
Journal of Medical Internet Research, 26, e54538 (2024) - JCR Q1; JIF 8.2; 96th percentile, rank 8/194 in Health Care Sciences & Services (2026 JCR)

INTERNATIONAL CONFERENCE PAPERS

Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot
Bogyeom Park and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

How Self-Disclosing Chatbots Influence Student Engagement, Assessment Accuracy, and Self-Reflection in Academic Stress Assessment
Minyoung Park, **Bogyeom Park**, and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

A Self-Determination Theory-Based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement
Hyerim Han, **Bogyeom Park**, and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment
Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo
CHI EA '24: Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (33.9% acceptance; 391/1,154)

Multimodal Machine Learning Model for MCI Detection Using EEG, MRI and VR Data
Mariem Kallel, **Bogyeom Park**, Kyoungwon Seo, and Seong-Eun Kim
2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

Advancing Mild Cognitive Impairment Detection: Integrating VR, MRI, and Neuropsychological Insights for Comprehensive Diagnosis
Bogyeom Park, Jinseok Park, Hojin Choi, Hokyoung Ryu, and Kyoungwon Seo
2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

Early Screening of Mild Cognitive Impairment Using Multimodal VR-EP-EEG-MRI (VEEM) Biomarkers via Machine Learning
Se Young Kim, **Bogyeom Park**, Dohyun Kim, Hojin Choi, Jinseok Park, Hokyoung Ryu, and Kyoungwon Seo
2024 International Conference on Electronics, Information, and Communication (ICEIC), pp. 1-4

KOREAN CONFERENCE PAPERS

From Teacher Needs to Agentic AI: Designing and Validating a Personalized Career Counseling System
Bogyeom Park, Mina Yoo, Dongkuk Lee, Mi-ae Choi, Seona Park, So Young Jo, and Kyoungwon Seo
Proceedings of HCI Korea 2026 - Oral Presentation

Counterfactual vs. Prefactual: Two Narrative AIs Improve Causability for Health Data by Different Mechanisms
Hyobin Park (now **Bogyeom Park**) and Kyoungwon Seo
Proceedings of HCI Korea 2023, pp. 828-835

RESEARCH EXPERIENCE

Human-centered Artificial Intelligence (HAI) Lab, SeoulTech Seoul, South Korea
Graduate Researcher (Advisor: Kyoungwon Seo) *Mar. 2023 - Present*
Undergraduate Researcher *Mar. 2021 - Feb. 2023*

Amounts are each project's total award, held by my advisor as principal investigator; the role given is the one I held on the research team.

Usability Evaluation of a Compound AI Tutoring System
Lead Graduate Researcher | Tutorus Labs | KRW 30M *Mar. 2026 - Aug. 2026*

- Designed the evaluation framework and wrote the utterance-level ICAP coding manual, then built the classifier that applies it
- Ran the sessions with school and first-year university students, and built the analytics dashboard linking engagement to learning outcome
- Produced a coded dialogue corpus and the project's final report; journal manuscript in preparation

Agentic AI for Personalized Career, Admission, and Counseling Support
Lead Graduate Researcher | Korea Education & Research Information Service | KRW 50M *Jun. 2025 - Dec. 2025*

- Ran the teacher focus groups, designed the multi-agent workflow, and ran two rounds of expert validation and a Delphi study
- Piloted the system with 30 teachers and 150 students; experts rated the design direction 4.37/5 and automatic record ingestion 5.00/5
- Produced KERIS Research Report RR 2025-3, an invention disclosure, and a first-authored HCI Korea 2026 paper

AI Copilot for Teacher-Augmented, Competence-Adaptive Learning
Lead Graduate Researcher | Institute for Information & Communications Technology Planning & Evaluation | KRW 1B *Jul. 2023 - Dec. 2025*

- Designed and evaluated a multi-agent LLM debate chatbot that draws out a student's argument and scores critical thinking from the exchange
- Its evaluator agents matched human raters at an intraclass correlation of 0.78, agreeing within one point 97.37% of the time (CHI EA 2025, first-authored)
- Co-developed a self-determination-theory career counseling chatbot

AI Agents for Improving Teachers' Instructional Design
Lead Graduate Researcher | Korea Education & Research Information Service — *2025*

- Authored the final report, designed the system prompts, and drafted the interface
- Produced a practitioner guide to collaborative instructional-design agents

VR-EP-EEG-MRI Digital Biomarker Basic Research Laboratory
Lead Graduate Researcher | Ministry of Science and ICT | KRW 1.2B — *Jun. 2021 - Feb. 2024*

- Ran the VR data collection on site at Hanyang University Guri Hospital, with patients rather than a convenience sample
- Analyzed the VEEM multimodal dataset - hand movement, gaze, evoked potentials, EEG and MRI - including the VR-MRI-SNSB cohort of 54 participants
- The dataset behind four of my papers; a filed patent was submitted as a project deliverable

Digital Biomarkers for Early Dementia Diagnosis from VR Daily-Living Data
Lead Graduate Researcher | Ministry of Science and ICT | KRW 550M — *Mar. 2021 - Feb. 2024*

- Built the virtual kiosk task in Unity - the six-step daily-living scenario the study's VR biomarkers are derived from
- Led the multimodal model, deriving hand-movement and gaze features and reading them together with MRI-derived measures
- Separated 22 healthy controls from 32 patients at 94.4% accuracy, higher than VR or MRI alone (JMIR 2024, first-authored)

TEACHING & MENTORING

2026 AX Academy Big Data Boot Camp — *May 2026 - Aug 2026*
Tutor, Hyundai Motor Group

- Mentored nine participants from across the company, each developing an individual big data project from proposal to final deliverable
- Advised on analysis design and revised project plans with participants through the boot camp's project-based sprints

Deep Learning — *Fall 2023*
Teaching Assistant, Seoul National University of Science and Technology

- Designed a final project using CNN-based models to predict drivers' physical and cognitive states from image data
- Supported lectures, advised student projects, and graded assignments and examinations

SKILLS

- **Research Methods -** experimental design, usability evaluation, focus group interviews, Delphi and expert validation, school and field pilots, clinical data collection with patients, utterance-level dialogue coding, inter-rater agreement, statistical analysis
- **Machine Learning -** PyTorch, scikit-learn, Hugging Face, feature engineering, multimodal fusion, transfer learning, classifier development
- **LLM Systems -** OpenAI and Anthropic APIs, multi-agent pipelines, RAG, prompt engineering and evaluation, LLM-as-a-judge
- **Programming and Tooling -** Python, C/C#, Docker, Streamlit, Git, SPSS
- **VR and Interactive Systems -** Unity, VR task development, interaction logging and eye tracking
- **Design -** HCI theory, UX design, interaction prototyping, usability testing

PATENT

Explainable AI-Based System for Early Diagnosis and Prognosis Prediction of Alzheimer's Disease Using VR Biomarkers, and Method Thereof (Korean Patent Application No. 10-2023-0105821, filed Aug. 2023)

ACADEMIC SERVICE

- Student Volunteer, AIED 2026
- Publicity Manager, SeoulTech Human-centered Artificial Intelligence Lab